

Project Title: School Bus Tracking & Parent Notification App

1. Problem Statement

Many schools still use manual methods such as phone calls, WhatsApp messages, or fixed schedules to inform parents about school bus arrivals and departures. These methods do not provide real-time updates, making it difficult for parents to know the exact location of the bus. Delays caused by traffic, weather, or other unforeseen situations can increase parents' anxiety and lead to students waiting for long periods at bus stops. There is a need for a reliable system that provides live bus tracking and instant notifications to improve communication, student safety, and the overall efficiency of school transportation.

2. Objectives

- Develop a real-time GPS-based school bus tracking system.
 - Provide parents with live bus location and estimated arrival time (ETA).
 - Send instant notifications for bus departure, arrival, delays, and trip completion.
 - Improve student safety by allowing parents to monitor bus movements.
 - Enable school administrators to manage buses, routes, drivers, and students efficiently.
 - Reduce communication gaps between schools, parents, and drivers.
 - Provide an easy-to-use and secure application for all users.
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3. Scope

The School Bus Tracking & Parent Notification App is designed to improve the management of school transportation through real-time monitoring and communication. The system allows administrators to manage buses, routes, drivers, and student details. Drivers can share their live GPS location during trips, while parents can view the bus location on a map and receive timely notifications about important events such as departure, arrival, delays, and trip completion. The application is intended for schools with one or more buses and can be expanded in the future with additional features such as attendance tracking, emergency alerts, and AI-based route optimization.

4. Expected Outcomes

- A functional application that displays the live location of school buses.
- Parents receive accurate and timely notifications regarding bus movements.
- Reduced waiting time for students and parents at bus stops.
- Improved communication between school administrators, drivers, and parents.
- Enhanced student safety through continuous monitoring of bus trips.
- Better management of school transportation with centralized data and reports.
- Increased trust and satisfaction among parents due to greater transparency and reliability in the school transportation system.